

Final Project

Detailed Testing Procedures: Nmap and Wireshark

Nmap Testing Phase

Pre-Testing Setup

```
bash
# Update Parrot OS
sudo apt update && sudo apt upgrade

# Ensure Nmap is installed (should be pre-installed)
nmap --version

# Check network configuration
ip addr show
route -n
```

Phase 1: Network Discovery

1.1 Host Discovery

```
bash
# Ping sweep to discover live hosts
nmap -sn 192.168.1.0/24

# Save results to file
nmap -sn 192.168.1.0/24 > host_discovery.txt
```

Document Results:

- Number of hosts discovered: 29
- IP addresses found: 29
- Response times: 2.7 sec

1.2 ARP Discovery (Local Network)

bash

ARP ping scan

`nmap -sn -PR 192.168.1.0/24`

Compare with system ARP table

`arp -a`

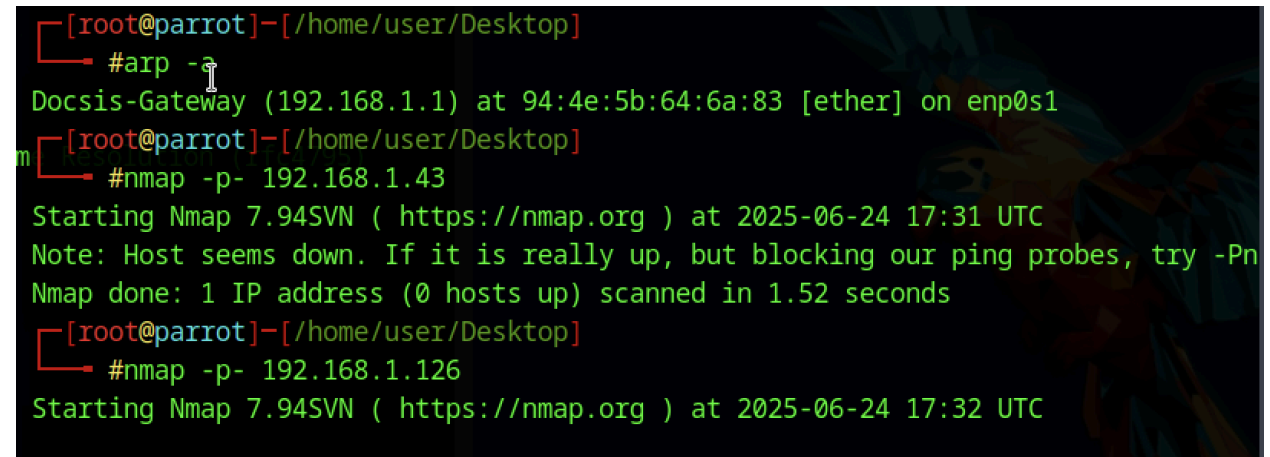
Phase 2: Port Scanning

2.1 Basic Port Scans

bash

Scan all 65535 ports

`nmap -p- [TARGET_IP]`

A terminal window screenshot with a dark background and a parrot illustration on the right. The terminal shows the following commands and output:

```
[root@parrot]-[/home/user/Desktop]
#arp -a
Docsis-Gateway (192.168.1.1) at 94:4e:5b:64:6a:83 [ether] on enp0s1
m [root@parrot]-[/home/user/Desktop]
#nmap -p- 192.168.1.43
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-24 17:31 UTC
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.52 seconds
[root@parrot]-[/home/user/Desktop]
#nmap -p- 192.168.1.126
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-24 17:32 UTC
```

Phase 3: Service and OS Detection

3.1 Service Version Detection

bash

Basic service detection

`V [TARGET_IP]`

```
[root@parrot]-[/home/user]
#nmap -sV 192.168.1.126
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-24 17:51 UTC
Nmap scan report for Moksha (192.168.1.126)
Host is up (0.040s latency).
All 1000 scanned ports on Moksha (192.168.1.126) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 28:D0:43:DE:6F:18 (Unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 42.31 seconds
```

3.2 Operating System Detection

bash

OS detection

nmap -O [TARGET_IP]

```
[root@parrot]-[/home/user]
#nmap -o 192.168.1.126
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-24 17:53 UTC
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.04 seconds
```

4.2 NSE (Nmap Scripting Engine)

bash

Run default scripts

nmap -sC [TARGET_IP]

```
[root@parrot]-[/home/user]
#nmap -sC 192.168.1.126
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-24 17:53 UTC
Nmap scan report for Moksha (192.168.1.126)
Host is up (0.090s latency).
All 1000 scanned ports on Moksha (192.168.1.126) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 28:D0:43:DE:6F:18 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 96.99 seconds
```

Run vulnerability scripts

nmap --script vuln [TARGET_IP]

```
[root@parrot]-[/home/user]
#nmap --script vuln 192.168.1.126
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-24 17:59 UTC
Pre-scan script results:
| broadcast-avahi-dos:
|   Discovered hosts:
|     224.0.0.251
|   After NULL UDP avahi packet DoS (CVE-2011-1002).
|_ Hosts are all up (not vulnerable).
Nmap scan report for Moksha (192.168.1.126)
Host is up (0.056s latency).
All 1000 scanned ports on Moksha (192.168.1.126) are in ignored states
.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 28:D0:43:DE:6F:18 (Unknown)
Nmap done: 1 IP address (1 host up) scanned in 93.41 seconds
```

Run specific script categories

```
nmap --script auth [TARGET_IP]
```

```
[root@parrot]-[/home/user]
#nmap --script auth 192.168.1.126
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-24 18:02 UTC
Nmap scan report for Moksha (192.168.1.126)
Host is up (0.12s latency).
All 1000 scanned ports on Moksha (192.168.1.126) are in ignored states
.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 28:D0:43:DE:6F:18 (Unknown)
Nmap done: 1 IP address (1 host up) scanned in 124.14 seconds
```

```
nmap --script brute [TARGET_IP]
```

```
[root@parrot]-[/home/user]
#nmap --script brute 192.168.1.126
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-24 18:05 UTC
Nmap scan report for Moksha (192.168.1.126)
Host is up (0.040s latency).
All 1000 scanned ports on Moksha (192.168.1.126) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 28:D0:43:DE:6F:18 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 47.08 seconds
```

```
nmap --script discovery [TARGET_IP]
```

Wireshark Analysis Phase

Phase 1: Traffic Capture Setup

1.1 Interface Selection and Configuration

bash

```
# List available interfaces
```

```
ip link show
```

```
# or
```

```
ifconfig -a
```

```
# Check interface status
```

```
sudo wireshark
```

```
# Select appropriate interface (usually eth0, wlan0, or similar)
```

1.2 Capture Filters (Set before starting capture)

```
# Capture only traffic to/from specific host
```

```
host 192.168.1.222
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.1.222	192.168.1.1	DNS	81	Standard query 0x0b4e A start.parrotsec.org
2	0.000038334	192.168.1.222	192.168.1.1	DNS	81	Standard query 0x684c AAAA start.parrotsec.org
3	0.156327907	UbeeInte_64:6a:83		ARP	62	Who has 192.168.1.222? Tell 192.168.1.1
4	0.156349466	1a:8e:d5:f6:e1:07		ARP	44	192.168.1.222 is at 1a:8e:d5:f6:e1:07
5	0.162160966	192.168.1.1	192.168.1.222	DNS	171	Standard query response 0x0b4e A start.parrotsec.org CNAME mordor.backbon...
6	0.162161341	192.168.1.1	192.168.1.222	DNS	183	Standard query response 0x684c AAAA start.parrotsec.org CNAME mordor.back...
7	0.188304547	192.168.1.222	192.168.1.1	DNS	97	Standard query 0x0290 A content-signature-2.cdn.mozilla.net
8	0.188324839	192.168.1.222	192.168.1.1	DNS	97	Standard query 0x9093 AAAA content-signature-2.cdn.mozilla.net
9	0.206073388	192.168.1.1	192.168.1.222	DNS	183	Standard query response 0x0290 A content-signature-2.cdn.mozilla.net CNAM...
10	0.206073721	192.168.1.1	192.168.1.222	DNS	195	Standard query response 0x9093 AAAA content-signature-2.cdn.mozilla.net C...
11	0.541445891	192.168.1.222	192.168.1.1	DNS	75	Standard query 0x407a A p.typekit.net
12	0.542430919	192.168.1.222	192.168.1.1	DNS	75	Standard query 0x6804 AAAA p.typekit.net
13	0.555197409	192.168.1.1	192.168.1.222	DNS	186	Standard query response 0x407a A p.typekit.net CNAME p.typekit.net-stls-v...
14	0.621183171	192.168.1.1	192.168.1.222	DNS	210	Standard query response 0x6804 AAAA p.typekit.net CNAME p.typekit.net-stl...
15	1.031559318	192.168.1.222	192.168.1.1	DNS	77	Standard query 0xba24 A use.typekit.net
16	1.031579027	192.168.1.222	192.168.1.1	DNS	77	Standard query 0x1926 AAAA use.typekit.net
17	1.053003014	192.168.1.1	192.168.1.222	DNS	185	Standard query response 0xba24 A use.typekit.net CNAME use-stls.adobe.com...

```
# Capture only TCP traffic
```

Tcp

No.	Time	Source	Destination	Protocol	Length	Info
17	4.796948633	2600:480a:3393:6900::	2600:1901:0:92a9::	TCP	88	34962 → 443 [ACK] Seq=1 Ack=1 Win=64896 Len=0 TSval=377614319 TSecr=98754...
18	4.797476240	2600:480a:3393:6900::	2600:1901:0:92a9::	TLSv1.2	304	Client Hello
19	4.811979181	2600:1901:0:92a9::	2600:480a:3393:6900::	TCP	88	443 → 34962 [ACK] Seq=1 Ack=217 Win=269824 Len=0 TSval=987540657 TSecr=37...
20	4.818967619	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	2504	Server Hello
21	4.819678669	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	693	Certificate, Server Key Exchange, Server Hello Done
22	4.819611246	2600:480a:3393:6900::	2600:1901:0:92a9::	TCP	88	34962 → 443 [ACK] Seq=217 Ack=2417 Win=68992 Len=0 TSval=377614342 TSecr=...
23	4.819108000	2600:480a:3393:6900::	2600:1901:0:92a9::	TCP	88	34962 → 443 [ACK] Seq=217 Ack=3022 Win=69504 Len=0 TSval=377614342 TSecr=...
24	4.822168220	2600:480a:3393:6900::	2600:1901:0:92a9::	TLSv1.2	181	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
25	4.839837552	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	399	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
26	4.839837761	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	157	Application Data
27	4.840367698	2600:480a:3393:6900::	2600:1901:0:92a9::	TCP	88	34962 → 443 [ACK] Seq=310 Ack=3402 Win=71808 Len=0 TSval=377614363 TSecr=...
28	4.840708425	2600:480a:3393:6900::	2600:1901:0:92a9::	TLSv1.2	187	Application Data
29	4.840728051	2600:480a:3393:6900::	2600:1901:0:92a9::	TLSv1.2	347	Application Data
30	4.840933351	2600:480a:3393:6900::	2600:1901:0:92a9::	TLSv1.2	126	Application Data
31	4.857574929	2600:1901:0:92a9::	2600:480a:3393:6900::	TCP	88	443 → 34962 [ACK] Seq=3402 Ack=706 Win=269568 Len=0 TSval=987540703 TSecr=...
32	4.857575096	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	126	Application Data
33	4.857575138	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	420	Application Data
34	4.857575179	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	1296	Application Data
35	4.857663100	2600:480a:3393:6900::	2600:1901:0:92a9::	TCP	88	34962 → 443 [ACK] Seq=706 Ack=4980 Win=71168 Len=0 TSval=377614380 TSecr=...
36	4.859218628	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	1296	Application Data
37	4.859218794	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	2504	Application Data, Application Data
38	4.859218836	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	713	Application Data
39	4.859218836	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	134	Application Data
40	4.859268297	2600:480a:3393:6900::	2600:1901:0:92a9::	TCP	88	34962 → 443 [ACK] Seq=706 Ack=9275 Win=67456 Len=0 TSval=377614382 TSecr=...
41	4.859621938	2600:480a:3393:6900::	2600:1901:0:92a9::	TLSv1.2	134	Application Data
42	4.873489141	2600:1901:0:92a9::	2600:480a:3393:6900::	TCP	88	443 → 34962 [ACK] Seq=9275 Ack=752 Win=269568 Len=0 TSval=987540721 TSecr=...

Capture HTTP and HTTPS

No.	Time	Source	Destination	Protocol	Length	Info
74	6.259938979	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
76	6.311657916	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
254	9.126110627	192.168.1.222	34.107.221.82	HTTP	386	GET /success.txt?ipv4 HTTP/1.1
261	9.140993249	2600:480a:3393:6900...	2600:1901:0:38d7::	HTTP	406	GET /success.txt?ipv6 HTTP/1.1
266	9.148863558	34.107.221.82	192.168.1.222	HTTP	284	HTTP/1.1 200 OK (text/plain)
270	9.165483424	2600:1901:0:38d7::	2600:480a:3393:6900...	HTTP	304	HTTP/1.1 200 OK (text/plain)
337	10.133441141	2600:480a:3393:6900...	2607:f8b0:4006:80b:...	HTTP	550	GET / HTTP/1.1
340	10.163977918	2607:f8b0:4006:80b:...	2600:480a:3393:6900...	HTTP	1053	HTTP/1.1 301 Moved Permanently (text/html)
370	10.319425675	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
377	10.371311987	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
406	10.425635073	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	524	Request
413	10.478103827	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1191	Response
537	10.950311513	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
543	10.999522245	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
638	11.447037745	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
674	11.499378074	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
713	11.543487037	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	524	Request
745	11.582560276	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
746	11.591853065	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1191	Response
769	11.626764700	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
771	11.635347874	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
783	11.658363689	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	524	Request
800	11.686492231	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
811	11.745001458	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1191	Response
2600	23.361656122	2600:480a:3393:6900...	2606:4700:4400::681...	OCSP	526	Request
2603	23.364913516	2600:480a:3393:6900...	2606:4700:4400::681...	OCSP	526	Request

port 80 or port 443

Phase 2: Synchronized Testing

2.1 Baseline Traffic Capture

bash

```
# Start Wireshark capture on target network interface
```

```
# Let it run for 5-10 minutes to establish baseline
```

```
# Document normal traffic patterns:
```

```
# - DNS queries
```

```
# - Regular HTTP/HTTPS
```

```
# - SSH sessions
```

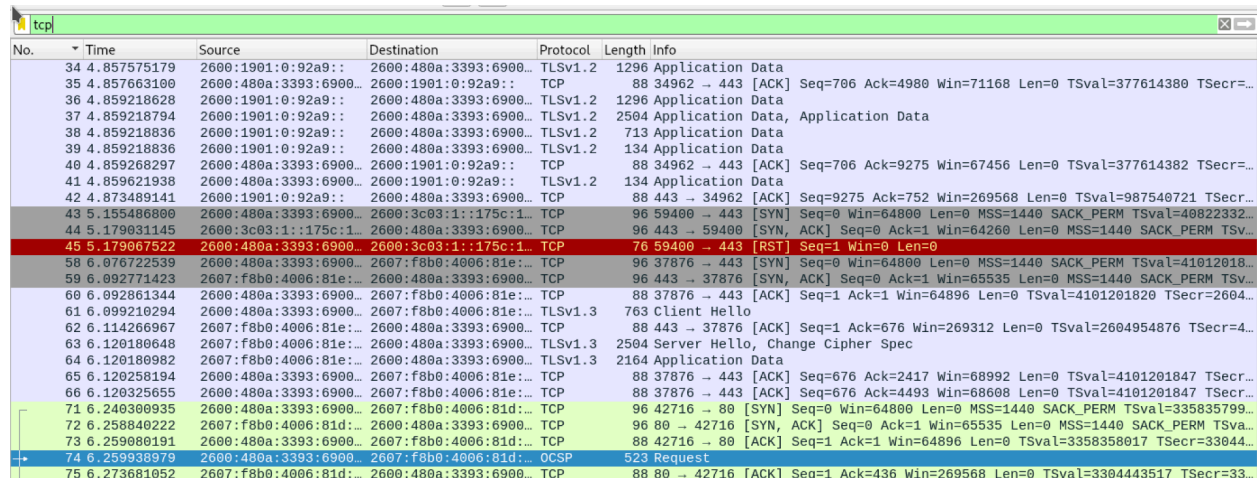
```
# - Any other legitimate traffic
```

Phase 3: Traffic Analysis Procedures

3.1 Display Filters (Use during analysis)

```
# Show only TCP traffic
```

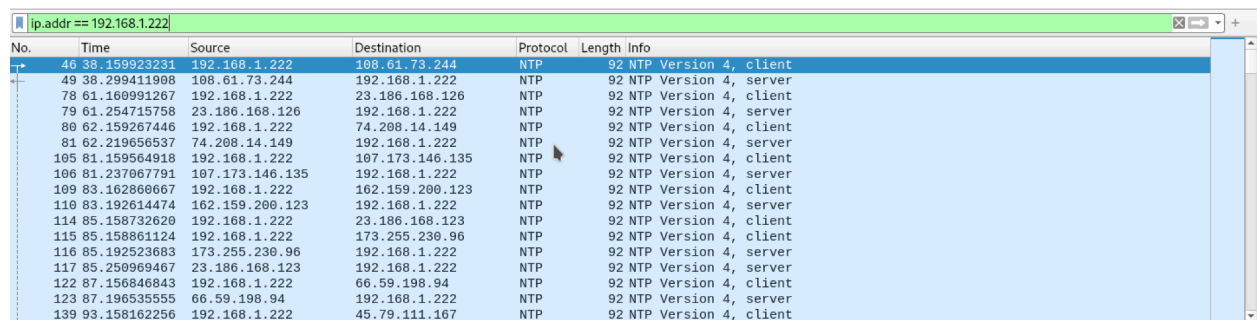
tcp



No.	Time	Source	Destination	Protocol	Length	Info
34	4.857575179	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	1296	Application Data
35	4.857663100	2600:480a:3393:6900::	2600:1901:0:92a9::	TCP	88	34962 → 443 [ACK] Seq=706 Ack=4980 Win=71168 Len=0 TSval=377614380 TSecr=...
36	4.859218628	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	1296	Application Data
37	4.859218794	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	2504	Application Data, Application Data
38	4.859218836	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	713	Application Data
39	4.859218836	2600:1901:0:92a9::	2600:480a:3393:6900::	TLSv1.2	134	Application Data
40	4.859268297	2600:480a:3393:6900::	2600:1901:0:92a9::	TCP	88	34962 → 443 [ACK] Seq=706 Ack=9275 Win=67456 Len=0 TSval=377614382 TSecr=...
41	4.859621938	2600:480a:3393:6900::	2600:1901:0:92a9::	TLSv1.2	134	Application Data
42	4.873489141	2600:1901:0:92a9::	2600:480a:3393:6900::	TCP	88	443 → 34962 [ACK] Seq=9275 Ack=752 Win=269568 Len=0 TSval=987540721 TSecr=...
43	5.155406000	2600:480a:3393:6900::	2600:3c03:1:175c:1::	TCP	96	59400 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=40822332...
44	5.179031145	2600:3c03:1:175c:1::	2600:480a:3393:6900::	TCP	96	443 → 59400 [SYN, ACK] Seq=0 Ack=1 Win=64260 Len=0 MSS=1440 SACK_PERM TSv...
45	5.179067522	2600:480a:3393:6900::	2600:3c03:1:175c:1::	TCP	76	59400 → 443 [RST] Seq=1 Win=0 Len=0
58	6.076722539	2600:480a:3393:6900::	2607:f8b0:4006:81e::	TCP	96	37876 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=41012018...
59	6.092771423	2607:f8b0:4006:81e::	2600:480a:3393:6900::	TCP	96	443 → 37876 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM TSv...
60	6.092861344	2600:480a:3393:6900::	2607:f8b0:4006:81e::	TCP	88	37876 → 443 [ACK] Seq=1 Ack=1 Win=64896 Len=0 TSval=4101201820 TSecr=2604...
61	6.099210294	2600:480a:3393:6900::	2607:f8b0:4006:81e::	TLSv1.3	763	Client Hello
62	6.114266967	2607:f8b0:4006:81e::	2600:480a:3393:6900::	TCP	88	443 → 37876 [ACK] Seq=1 Ack=676 Win=269312 Len=0 TSval=2604954876 TSecr=4...
63	6.120180648	2607:f8b0:4006:81e::	2600:480a:3393:6900::	TLSv1.3	2504	Server Hello, Change Cipher Spec
64	6.120180982	2607:f8b0:4006:81e::	2600:480a:3393:6900::	TLSv1.3	2164	Application Data
65	6.120258194	2600:480a:3393:6900::	2607:f8b0:4006:81e::	TCP	88	37876 → 443 [ACK] Seq=676 Ack=2417 Win=68992 Len=0 TSval=4101201847 TSecr=...
66	6.120325655	2600:480a:3393:6900::	2607:f8b0:4006:81e::	TCP	88	37876 → 443 [ACK] Seq=676 Ack=4493 Win=68608 Len=0 TSval=4101201847 TSecr=...
71	6.240300935	2600:480a:3393:6900::	2607:f8b0:4006:81d::	TCP	96	42716 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=335835799...
72	6.258840222	2607:f8b0:4006:81d::	2600:480a:3393:6900::	TCP	96	80 → 42716 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM TSva...
73	6.259080191	2600:480a:3393:6900::	2607:f8b0:4006:81d::	TCP	88	42716 → 80 [ACK] Seq=1 Ack=1 Win=64896 Len=0 TSval=3358358017 TSecr=33044...
74	6.259938979	2600:480a:3393:6900::	2607:f8b0:4006:81d::	OCSP	523	Request
75	6.273681052	2607:f8b0:4006:81d::	2600:480a:3393:6900::	TCP	88	80 → 42716 [ACK] Seq=1 Ack=436 Win=269568 Len=0 TSval=3304443517 TSecr=33...

```
# Show only packets to/from specific IP
```

```
ip.addr == 192.168.1.222
```



No.	Time	Source	Destination	Protocol	Length	Info
46	38.159923231	192.168.1.222	198.61.73.244	NTP	92	NTP Version 4, client
49	38.299411908	198.61.73.244	192.168.1.222	NTP	92	NTP Version 4, server
78	61.160991267	192.168.1.222	23.186.168.126	NTP	92	NTP Version 4, client
79	61.254715758	23.186.168.126	192.168.1.222	NTP	92	NTP Version 4, server
80	62.159267446	192.168.1.222	74.208.14.149	NTP	92	NTP Version 4, client
81	62.219656537	74.208.14.149	192.168.1.222	NTP	92	NTP Version 4, server
105	81.159564918	192.168.1.222	107.173.146.135	NTP	92	NTP Version 4, client
106	81.237067791	107.173.146.135	192.168.1.222	NTP	92	NTP Version 4, server
109	83.162860667	192.168.1.222	162.159.200.123	NTP	92	NTP Version 4, client
110	83.192614474	162.159.200.123	192.168.1.222	NTP	92	NTP Version 4, server
114	85.158732620	192.168.1.222	23.186.168.123	NTP	92	NTP Version 4, client
115	85.158861124	192.168.1.222	173.255.230.96	NTP	92	NTP Version 4, client
116	85.192523683	173.255.230.96	192.168.1.222	NTP	92	NTP Version 4, server
117	85.250969467	23.186.168.123	192.168.1.222	NTP	92	NTP Version 4, server
122	87.156846843	192.168.1.222	66.59.198.94	NTP	92	NTP Version 4, client
123	87.196535555	66.59.198.94	192.168.1.222	NTP	92	NTP Version 4, server
139	93.158162256	192.168.1.222	45.79.111.167	NTP	92	NTP Version 4, client

```
# Show HTTP traffic
```

http

No.	Time	Source	Destination	Protocol	Length	Info
74	6.259938979	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
76	6.311657916	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
254	9.126110627	192.168.1.222	34.107.221.82	HTTP	386	GET /success.txt?ipv4 HTTP/1.1
261	9.148993249	2600:480a:3393:6900...	2600:1901:0:38d7::	HTTP	406	GET /success.txt?ipv6 HTTP/1.1
266	9.148863558	34.107.221.82	192.168.1.222	HTTP	284	HTTP/1.1 200 OK (text/plain)
279	9.165483424	2600:1901:0:38d7::	2600:480a:3393:6900...	HTTP	304	HTTP/1.1 200 OK (text/plain)
337	10.133441141	2600:480a:3393:6900...	2607:f8b0:4006:80b:...	HTTP	550	GET / HTTP/1.1
340	10.163977918	2607:f8b0:4006:80b:...	2600:480a:3393:6900...	HTTP	1053	HTTP/1.1 301 Moved Permanently (text/html)
370	10.319425675	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
377	10.371311987	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
406	10.425635073	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	524	Request
413	10.478103827	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1191	Response
537	10.950311513	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
543	10.995222245	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
638	11.447037745	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	523	Request
674	11.499378074	2607:f8b0:4006:81d:...	2600:480a:3393:6900...	OCSP	1190	Response
713	11.543487037	2600:480a:3393:6900...	2607:f8b0:4006:81d:...	OCSP	524	Request

Show DNS queries

dns

No.	Time	Source	Destination	Protocol	Length	Info
11	4.779012498	192.168.1.1	192.168.1.222	DNS	195	Standard query response 0x2b2a AAAA content-signature-2.cdn.mozilla.net C...
12	4.779012956	192.168.1.1	192.168.1.222	DNS	183	Standard query response 0x3428 A content-signature-2.cdn.mozilla.net CNAM...
14	4.782794042	192.168.1.1	192.168.1.222	DNS	171	Standard query response 0x37e1 A start.parrotsec.org CNAME mordor.backbon...
15	4.782794250	192.168.1.1	192.168.1.222	DNS	183	Standard query response 0xd1e0 AAAA start.parrotsec.org CNAME mordor.back...
48	5.661145045	192.168.1.222	192.168.1.1	DNS	77	Standard query 0x4734 A use.typekit.net
49	5.661158379	192.168.1.222	192.168.1.1	DNS	77	Standard query 0x3135 AAAA use.typekit.net
50	5.682798888	192.168.1.1	192.168.1.222	DNS	209	Standard query response 0x3135 AAAA use.typekit.net CNAME use-stls.adobe...
51	5.682799430	192.168.1.1	192.168.1.222	DNS	185	Standard query response 0x4734 A use.typekit.net CNAME use-stls.adobe.com...
53	6.049117722	192.168.1.222	192.168.1.1	DNS	89	Standard query 0x2785 A safebrowsing.googleapis.com
54	6.049132722	192.168.1.222	192.168.1.1	DNS	89	Standard query 0x3187 AAAA safebrowsing.googleapis.com
56	6.075703869	192.168.1.1	192.168.1.222	DNS	105	Standard query response 0x2785 A safebrowsing.googleapis.com A 142.250.17...
57	6.075704160	192.168.1.1	192.168.1.222	DNS	117	Standard query response 0x3187 AAAA safebrowsing.googleapis.com AAAA 2607...
67	6.218050148	192.168.1.222	192.168.1.1	DNS	72	Standard query 0xdfa2 A o.pki.goog
68	6.218084525	192.168.1.222	192.168.1.1	DNS	72	Standard query 0x2ba0 AAAA o.pki.goog
69	6.239912959	192.168.1.1	192.168.1.222	DNS	123	Standard query response 0xdfa2 A o.pki.goog CNAME pki-goog.l.google.com A...
70	6.239913251	192.168.1.1	192.168.1.222	DNS	135	Standard query response 0x2ba0 AAAA o.pki.goog CNAME pki-goog.l.google.co...
106	6.453420903	192.168.1.222	192.168.1.1	DNS	87	Standard query 0x5018 A push.services.mozilla.com

3.2 Statistical Analysis

Statistics Menu → Protocol Hierarchy

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
▼ Frame	100.0	46	100.0	4136	1609	0	0	0
▼ Linux cooked-mode capture	100.0	46	17.8	736	286	0	0	0
▼ Internet Protocol Version 6	69.6	32	30.9	1280	498	0	0	0
Transmission Control Protocol	69.6	32	30.9	1280	498	32	1280	498
▼ Internet Protocol Version 4	30.4	14	6.8	280	108	0	0	0
Transmission Control Protocol	30.4	14	13.5	560	217	14	560	217

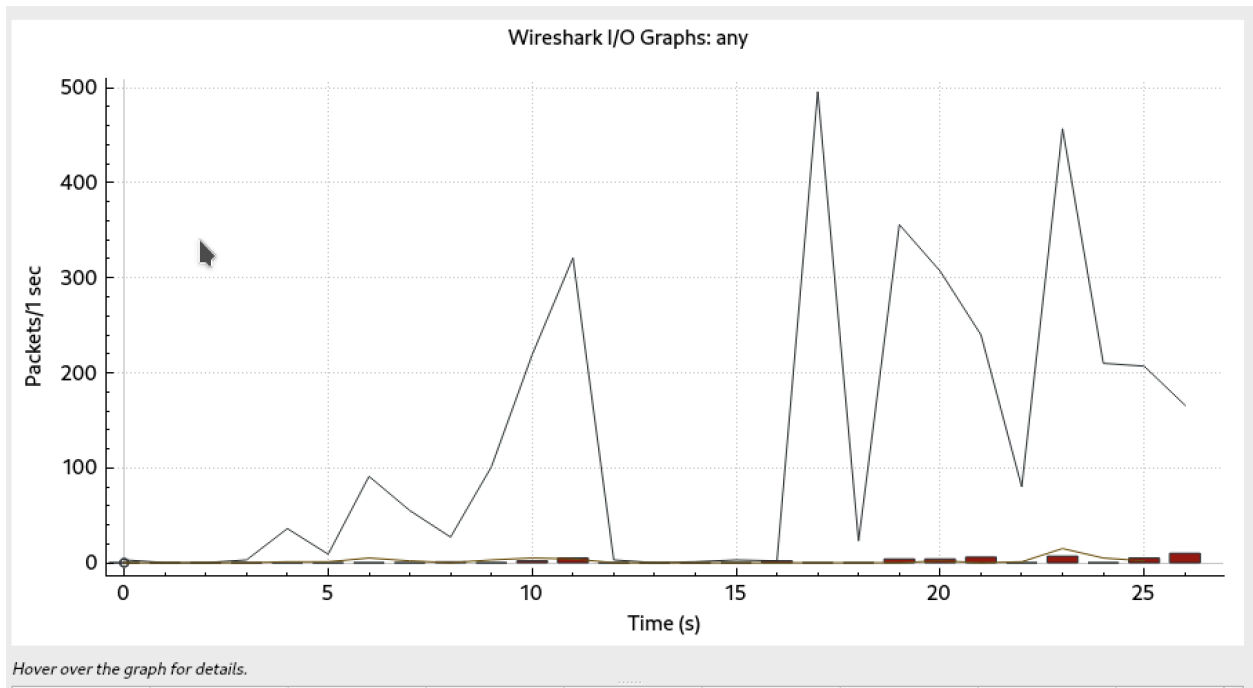
Statistics Menu → Conversations

Ethernet	IPv4 · 6	IPv6 · 18	TCP · 46	UDP							
Address A	Port A Address B		Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Pac		
192.168.1.222	43314	34.107.221.82	80	1	76 bytes	9	12	8.33%			
192.168.1.222	45782	34.107.243.93	443	1	76 bytes	4	23	4.35%			
192.168.1.222	45796	34.107.243.93	443	1	76 bytes	5	25	4.00%			
192.168.1.222	51002	35.196.174.192	443	1	76 bytes	26	30	3.33%			
192.168.1.222	51018	35.196.174.192	443	1	76 bytes	36	31	3.23%			
192.168.1.222	53022	35.243.152.122	443	1	76 bytes	27	32	3.13%			
192.168.1.222	53030	35.243.152.122	443	1	76 bytes	35	31	3.23%			
192.168.1.222	54958	199.232.90.133	443	1	76 bytes	41	29	3.45%			
192.168.1.222	54974	199.232.90.133	443	1	76 bytes	42	53	1.89%			
192.168.1.222	54986	199.232.90.133	443	1	76 bytes	43	29	3.45%			
192.168.1.222	53248	205.234.175.175	443	1	76 bytes	24	50	2.00%			
192.168.1.222	53252	205.234.175.175	443	1	76 bytes	25	24	4.17%			
192.168.1.222	53256	205.234.175.175	443	1	76 bytes	28	33	3.03%			
192.168.1.222	53264	205.234.175.175	443	1	76 bytes	38	25	4.00%			
2600:480a:3393:6900::902b	34616	2600:1901:0:38d7::	80	1	96 bytes	10	12	8.33%			
2600:480a:3393:6900::902b	32784	2600:1901:0:712f::	443	1	96 bytes	11	31	3.23%			
2600:480a:3393:6900::902b	34962	2600:1901:0:92a9::	443	1	96 bytes	0	36	2.78%			
2600:480a:3393:6900::902b	53036	2600:1901:0:c47c::	443	1	96 bytes	8	59	1.69%			
2600:480a:3393:6900::902b	59400	2600:3c03:1::175c:17b1	443	1	96 bytes	1	3	33.33%			
2600:480a:3393:6900::902b	59404	2600:3c03:1::175c:17b1	443	1	96 bytes	6	3	33.33%			
2600:480a:3393:6900::902b	59420	2600:3c03:1::175c:17b1	443	1	96 bytes	7	3	33.33%			
2600:480a:3393:6900::902b	46328	2600:9000:266a:7c00:c:b42a:3740:93a1	443	1	96 bytes	29	141	0.71%			
2600:480a:3393:6900::902b	47864	2606:4700:10::6816:858	443	1	96 bytes	37	39	2.56%			
2600:480a:3393:6900::902b	47862	2606:4700:20::ac43:5289	443	1	96 bytes	23	31	3.23%			
2600:480a:3393:6900::902b	42200	2606:4700:4400::6812:26e9	80	1	96 bytes	30	10	10.00%			

Statistics Menu → Endpoints

Ethernet	IPv4 - 7	IPv6 - 19	TCP - 70	UDP							
Address	Port	Packets	Bytes	Total Packets	Percent Filtered	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes		
34.107.221.82	80	1	76 bytes	12	8.33%	0	0 bytes	1	76 bytes		
34.107.243.93	443	2	152 bytes	48	4.17%	0	0 bytes	2	152 bytes		
35.196.174.192	443	2	152 bytes	61	3.28%	0	0 bytes	2	152 bytes		
35.243.152.122	443	2	152 bytes	63	3.17%	0	0 bytes	2	152 bytes		
192.168.1.222	43314	1	76 bytes	12	8.33%	1	76 bytes	0	0 bytes		
192.168.1.222	45782	1	76 bytes	23	4.35%	1	76 bytes	0	0 bytes		
192.168.1.222	45796	1	76 bytes	25	4.00%	1	76 bytes	0	0 bytes		
192.168.1.222	51002	1	76 bytes	30	3.33%	1	76 bytes	0	0 bytes		
192.168.1.222	51018	1	76 bytes	31	3.23%	1	76 bytes	0	0 bytes		
192.168.1.222	53022	1	76 bytes	32	3.13%	1	76 bytes	0	0 bytes		
192.168.1.222	53030	1	76 bytes	31	3.23%	1	76 bytes	0	0 bytes		
192.168.1.222	53248	1	76 bytes	50	2.00%	1	76 bytes	0	0 bytes		
192.168.1.222	53252	1	76 bytes	24	4.17%	1	76 bytes	0	0 bytes		
192.168.1.222	53256	1	76 bytes	33	3.03%	1	76 bytes	0	0 bytes		
192.168.1.222	53264	1	76 bytes	25	4.00%	1	76 bytes	0	0 bytes		
192.168.1.222	54958	1	76 bytes	29	3.45%	1	76 bytes	0	0 bytes		
192.168.1.222	54974	1	76 bytes	53	1.89%	1	76 bytes	0	0 bytes		
192.168.1.222	54986	1	76 bytes	29	3.45%	1	76 bytes	0	0 bytes		
199.232.90.133	443	3	228 bytes	111	2.70%	0	0 bytes	3	228 bytes		
205.234.175.175	443	4	304 bytes	132	3.03%	0	0 bytes	4	304 bytes		
2600:1901:0:38d7::	80	1	96 bytes	12	8.33%	0	0 bytes	1	96 bytes		
2600:1901:0:712f::	443	1	96 bytes	31	3.23%	0	0 bytes	1	96 bytes		
2600:1901:0:92a9::	443	1	96 bytes	36	2.78%	0	0 bytes	1	96 bytes		
2600:1901:0:c47c::	443	1	96 bytes	59	1.69%	0	0 bytes	1	96 bytes		
2600:3c03:1::175c:17b1	443	3	288 bytes	9	33.33%	0	0 bytes	3	288 bytes		
2600:480a:3393:6900::902b	32784	1	96 bytes	31	3.23%	1	96 bytes	0	0 bytes		

Statistics Menu → I/O Graphs



Phase 4: Specific Attack Pattern Analysis

4.1 Port Scan Detection

- # Look for patterns indicating port scanning:
- # 1. Multiple SYN packets to different ports from same source
- # 2. Quick succession of connection attempts
- # 3. Many RST responses from target

Filter to see potential port scan:

`tcp.flags.syn == 1 and tcp.flags.ack == 0`

No.	Time	Source	Destination	Protocol	Length	Info
13	4.781401896	2600:480a:3393:6900::	2600:1901:0:92a9::	TCP	96	34962 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=37761430...
43	5.155486800	2600:480a:3393:6900::	2600:3c03:1:175c:1::	TCP	96	59400 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=40822332...
58	6.076722539	2600:480a:3393:6900::	2607:f8b0:4006:81e::	TCP	96	37876 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=41012018...
71	6.240300935	2600:480a:3393:6900::	2607:f8b0:4006:81d::	TCP	96	42716 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=335835799...
110	6.499421434	192.168.1.222	34.107.243.93	TCP	76	45782 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=42933333...
124	6.588444914	192.168.1.222	34.107.243.93	TCP	76	45796 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=42933334...
139	6.936097649	2600:480a:3393:6900::	2600:3c03:1:175c:1::	TCP	96	59404 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=40822350...
145	7.163563225	2600:480a:3393:6900::	2600:3c03:1:175c:1::	TCP	96	59420 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=40822352...
152	7.226310527	2600:480a:3393:6900::	2600:1901:0:c47c::	TCP	96	53036 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=24624217...
247	9.110910241	192.168.1.222	34.107.221.82	TCP	76	43314 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=253424592...
255	9.126523062	2600:480a:3393:6900::	2600:1901:0:38d7::	TCP	96	34616 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=385882259...
293	9.387107805	2600:480a:3393:6900::	2600:1901:0:712f::	TCP	96	32784 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=23460458...
334	10.111749175	2600:480a:3393:6900::	2607:f8b0:4006:80b::	TCP	96	34130 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=108490766...
367	10.307179755	2600:480a:3393:6900::	2607:f8b0:4006:81d::	TCP	96	42732 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=335836206...
375	10.357622335	2600:480a:3393:6900::	2607:f8b0:4006:820::	TCP	96	44722 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=30978226...
401	10.411014172	2600:480a:3393:6900::	2607:f8b0:4006:81d::	TCP	96	42746 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=335836216...
476	10.749294480	2600:480a:3393:6900::	2607:f8b0:4006:81e::	TCP	96	37878 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=41012064...
659	11.488789103	2600:480a:3393:6900::	2607:f8b0:4006:822::	TCP	96	38964 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=17184986...
660	11.488859815	2600:480a:3393:6900::	2607:f8b0:4006:822::	TCP	96	38970 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=17184986...
732	11.557009101	2600:480a:3393:6900::	2607:f8b0:4006:80e::	TCP	96	43162 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=24470921...
763	11.612424771	2600:480a:3393:6900::	2607:f8b0:4006:81d::	TCP	96	50006 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=335836337...
1886	20.856545400	2600:480a:3393:6900::	2607:f8b0:4006:81c::	TCP	96	35624 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=11516269...
1902	20.874230856	2600:480a:3393:6900::	2607:f8b0:4006:81c::	TCP	96	35626 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=11516269...
2342	22.753674263	2600:480a:3393:6900::	2606:4700:20::ac43::	TCP	96	47862 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=11125608...
2428	23.270166468	192.168.1.222	205.234.175.175	TCP	76	53248 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=32543490...
2431	23.271075216	192.168.1.222	205.234.175.175	TCP	76	53252 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=32543490...

4.2 Service Enumeration Detection

Look for:

- # 1. Banner grabbing attempts
- # 2. Service-specific probes
- # 3. Version detection traffic

Filter for service detection:

tcp.len > 0 and tcp.flags.push == 1

No.	Time	Source	Destination	Protocol	Length	Info
18	4.797476240	2600:480a:3393:6900...	2600:1901:0:92a9::	TLSv1.2	304	Client Hello
20	4.818967619	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	2504	Server Hello
21	4.818967869	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	693	Certificate, Server Key Exchange, Server Hello Done
24	4.822168220	2600:480a:3393:6900...	2600:1901:0:92a9::	TLSv1.2	181	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
25	4.839837552	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	399	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
26	4.839837761	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	157	Application Data
28	4.840708425	2600:480a:3393:6900...	2600:1901:0:92a9::	TLSv1.2	187	Application Data
29	4.840728051	2600:480a:3393:6900...	2600:1901:0:92a9::	TLSv1.2	347	Application Data
30	4.840933351	2600:480a:3393:6900...	2600:1901:0:92a9::	TLSv1.2	126	Application Data
32	4.857575096	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	126	Application Data
33	4.857575138	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	420	Application Data
34	4.857575179	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	1296	Application Data
36	4.859218628	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	1296	Application Data
37	4.859218794	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	2504	Application Data, Application Data
38	4.859218836	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	713	Application Data
39	4.859218836	2600:1901:0:92a9::	2600:480a:3393:6900...	TLSv1.2	134	Application Data
41	4.859621938	2600:480a:3393:6900...	2600:1901:0:92a9::	TLSv1.2	134	Application Data
61	6.099210294	2600:480a:3393:6900...	2607:f8b0:4006:81e::	TLSv1.3	763	Client Hello
63	6.120180648	2607:f8b0:4006:81e::	2600:480a:3393:6900...	TLSv1.3	2504	Server Hello, Change Cipher Spec
64	6.120180982	2607:f8b0:4006:81e::	2600:480a:3393:6900...	TLSv1.3	2164	Application Data
74	6.259938979	2600:480a:3393:6900...	2607:f8b0:4006:81d::	OCSP	523	Request
76	6.311657916	2607:f8b0:4006:81d::	2600:480a:3393:6900...	OCSP	1190	Response
78	6.323864378	2600:480a:3393:6900...	2607:f8b0:4006:81e::	TLSv1.3	152	Change Cipher Spec, Application Data
79	6.324879882	2600:480a:3393:6900...	2607:f8b0:4006:81e::	TLSv1.3	180	Application Data
80	6.324899383	2600:480a:3393:6900...	2607:f8b0:4006:81e::	TLSv1.3	723	Application Data
82	6.345493929	2607:f8b0:4006:81e::	2600:480a:3393:6900...	TLSv1.3	702	Application Data, Application Data